Forecasting Weekly Depression Incidence in Mexico: A Multi-Model Framework with Auditable Per-Series Model Selection

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Abstract. Weekly forecasting of psychiatric demand is hard because surveillance signals are heterogeneous across regions, sex strata, and time, so no single forecasting paradigm performs uniformly well. Rather than imposing one algorithm across an entire surveillance system, we present a deterministic, auditable framework that selects, for each series, the model best supported by its own evidence. The framework draws on a library of forecasters with deliberately different inductive assumptions—a structural additive decomposition, a probabilistic recurrent network, and two tree-based hybrids—so that the candidate pool spans diverse temporal representations. A transparent rule then selects one model per series based on cross-validated accuracy, can reassign low-incidence series to a regional model, and records every decision in a justified selection log. We instantiate it for weekly depression incidence (CIE-10 F32) across Mexico's 32 federal entities, using public bulletins of the National Epidemiological Surveillance System (SINAVE), and validate it twice: by rolling-origin cross-validation over a decade of weekly observations and a prospective out-of-sample assessment against the 2026 bulletins released after the training cutoff. A leave-one-model-out ablation shows that the candidate pool is not what carries the result, and that the cross-validated ranking does not transport to the prospective window. The contribution is therefore not that several models beat one, but that an auditable mechanism detects when a frozen assignment stops transporting and revises it using only prior observations. The framework is reproducible and extends to conditions with heterogeneous behavior.

Keywords: Depression · Time series forecasting · Probabilistic recurrent networks · Multi-model forecasting · Epidemiological surveillance · Model selection.

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1 Introduction

Depression is a leading cause of disability worldwide and imposes a major burden on public health systems [25]. In Mexico, it affects an estimated 8–10 million people, and the National Epidemiological Surveillance System (SINAVE) records about 150,000 new cases of depressive episode (CIE-10 F32) per year [11]. Despite the routine availability of weekly case counts, psychiatric-service planning remains largely *reactive*: capacity is adjusted only after demand surges are already visible in the bulletin, producing recurrent saturation windows during the February–May period.

Accurate weekly federal-entity forecasts would make this planning *proactive*: a reliable 52-week-ahead trajectory would allow authorities to pre-position staff before seasonal peaks, size programs by state, and time help-seeking campaigns for quieter months. The problem is challenging precisely where forecasts are most consequential. The depression signal is highly heterogeneous: case loads differ by orders of magnitude across entities, are sharply skewed by sex, and range from smooth metropolitan series to volatile low-volume states in a panel marked by a COVID-19 regime change and a post-pandemic level shift. No single paradigm accommodates all of these at once: structural decompositions excel on smooth aggregates but lag on noisy series, whereas representation-learning models share structure across series but can overreact on smooth aggregates. Deployment also raises a trust problem: a model chosen on historical cross-validation can quietly lose its edge once the live signal shifts, so an institution needs a selection it can audit and revise, not a one-off leaderboard winner.

We address this problem with a **multi-model forecasting framework**, referred to throughout as *the proposed framework*. Rather than committing to one model family, it trains four forecasters with complementary inductive assumptions for each series and selects one model per series through a deterministic, auditable rule, so that the paradigm best supported by a given series produces that series' forecast. Although evaluated here on depression, the framework is designed for epidemiological surveillance in general: choosing one model per series is a direct response to the heterogeneity that makes any single architecture unlikely to dominate across the many conditions a surveillance system monitors.

The main contributions of this study are threefold: (i) we present and empirically evaluate a multi-model forecasting framework for psychiatric surveillance that places structural, recurrent, and tree-based forecasters behind a single interface, enabling a head-to-head, per-series comparison under one rolling-origin protocol; (ii) we introduce a deterministic, auditable per-series model-selection strategy that ranks candidates on cross-validated symmetric error with scale-free and large-error tie-breakers and a transparent low-incidence regional fallback, producing a fully justified, revisable selection log—an auditable selection mechanism built for trustworthy deployment under distribution shift rather than a one-off leaderboard choice; and (iii) we provide a nationwide empirical characterization, over 111 series–stratum combinations and 626 weeks of SINAVE data, of the regional, demographic, and temporal heterogeneity of the depression signal and of the conditions under which each forecasting paradigm prevails. A central finding is that the cross-validated ranking is a *selection* signal, not an out-of-sample guarantee: per-series accuracy degrades on live 2026 data, a degradation the revisable rule absorbs by reassigning models as the signal accumulates, substantiated by the 2026 bulletins published since the training cutoff.

2 Foundations and Related Work

2.1 Epidemiological Time-Series Forecasting

Incidence forecasting has progressed from linear, series-by-series statistical models to global, representation-learning architectures that share parameters across many related series. The Box–Jenkins autoregressive integrated moving average (ARIMA) family [5] remains a standard baseline, but its linearity and stationarity assumptions limit performance on non-stationary public-health signals subject to regime changes. Exponential-smoothing state-space models (ETS) [17] offer flexible additive or multiplicative decompositions but treat each series independently. Prophet [26] popularized an additive decomposition with explicit trend, seasonality, and holiday components and is robust to missing data and outliers, making it a frequent default in public-health pipelines; however, its smooth additive structure can underfit the volatile low-count series typical of small federal entities. Recurrent neural networks (RNNs), particularly long short-term memory (LSTM) networks, have shown clear advantages for epidemiological series: Chimmula and Zhang [8] applied LSTMs to COVID-19 incidence in Canada, outperforming classical baselines over short horizons, although such data-hungry models can overreact on smooth, high-volume aggregates for which simpler decompositions may already suffice. For public-health surveillance specifically, collaborative forecasting efforts have shown that systematic, reproducible evaluation of probabilistic forecasts is feasible at a national scale [23,9]. More recent architectures push long-horizon accuracy further—for example, the hierarchical-interpolation network N-HiTS [6]—and large pretrained *foundation* models for time series, such as Chronos [3], forecast across many domains with little or no task-specific training; their behavior on the sparse, regime-shifting panels typical of epidemiological surveillance remains largely untested.

2.2 Foundations of Multi-Model Forecasting

The case for combining heterogeneous forecasters is well established. Salinas et al. [24] introduced DeepAR, a global probabilistic autoregressive RNN that *learns* across a panel of related series and emits a full predictive distribution at each step; recent surveys of deep learning for forecasting situate such global models within a broad design space [4,19,14]. Large-scale empirical evidence from the M4 and M5 forecasting competitions—open benchmarks spanning 100,000 cross-domain series and a large hierarchical retail-sales corpus, respectively—shows that machine-learning and hybrid methods are competitive with, and often outperform, classical univariate methods across heterogeneous collections, and that combinations are particularly robust [20,21], including automated ensembles tuned to weekly series [13]. Combining the explicit components of additive decompositions with the nonlinear capacity of gradient-boosted decision trees (GBDTs) has proven effective in retail and energy forecasting [15], and Prophet–neural hybrids have improved pandemic incidence forecasts [10]. Stacked generalization [28] formalizes the learning of a meta-model over heterogeneous base learners. These foundations motivate a design in which structural, statistical, and learning-based assumptions coexist and a principled rule decides between them per series.

Why these four candidates. The pool samples the design space above rather than enumerating it: one representative per family of inductive assumption, chosen so that the selection rule adjudicates between genuinely different beliefs about the signal rather than between variants of one. Prophet contributes an explicit structural decomposition of trend and annual seasonality; DeepAR contributes a global probabilistic model that pools strength across the panel and emits a predictive distribution; and the two tree-based hybrids contribute nonlinear responses to calendar and lag features that neither of the first two represents, one blended with a structural component and one stacked over heterogeneous experts. Two further candidates were implemented and evaluated under the identical protocol and then left out of the deployed pool: a tuned Auto-ARIMA with Fourier annual terms, as the classical baseline the family above motivates, and PatchTST, as a patching transformer representative of the recent long-horizon architectures. Both are reported in Table 1 rather than hidden: neither reached the accuracy of the retained candidates on this panel, and PatchTST in particular did not repay its tuning cost at the 111-series scale. Foundation models such as Chronos [3] were not evaluated here and remain open. The pool is therefore a documented sample of the space, not a claim of completeness—and Sect. 5.1 tests how much of the result actually depends on it.

2.3 Depression and Mental-Health Surveillance

The burden of depression in Mexico has been quantified through the Global Burden of Disease study [12], and the COVID-19 pandemic produced a measurable global increase in depressive-disorder burden [25]. Sex-related disparities in reported incidence are well established clinically [1]. To our knowledge, no prior work has applied a multi-model forecasting framework with auditable per-series model selection to the SINAVE depression panel at federal-entity granularity while also providing a prospective out-of-sample evaluation.

3 Materials and Methods

3.1 Data Characterization

The study uses weekly case counts of *depressive episode*, the category the WHO International Classification of Diseases, 10th revision (ICD-10) codes as **F32**: a

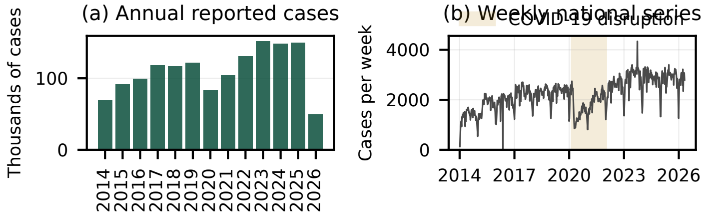

Fig. 1. Depression (CIE-10 F32) temporal distribution. (a) Annual reported cases 2014–2026 (2026 is year-to-date). (b) Weekly national series; the COVID-19 disruption window is highlighted in gold. The post-2022 level shift and the recurring February–May peaks motivate models that adapt across regimes.

first, single episode of depressed mood with loss of interest and reduced energy, graded mild, moderate, or severe, and distinct from recurrent depressive disorder (F33). SINAVE labels it with the Spanish acronym of the same classification, CIE-10. Counts are as reported in the SINAVE bulletin by the Dirección General de Epidemiología [11], Mexico's consolidated public health surveillance system. After calendar alignment under the SINAVE epidemiological-week convention,[3] the harmonized panel contains **626 weekly observations per series**, spanning 2014-W02 through 2025-W53. Epidemiological week 1 of 2014 was unavailable in the retrieved archive, so 2014 contributes 52 weeks; 2020 and 2025 each contain 53 epidemiological weeks, giving $626 = 10{\times}52{+}2{\times}53$. Records are disaggregated by sex (women and men) and aggregated into an overall total (general). We analyze the 32 federal entities, four socio-urban regions defined by population density and INEGI demographic indicators (*Metropolitana alta*, *Urbana media*, *Sur-Sureste vulnerable*, *Rural / dispersa*), and a national series, yielding $37 \times 3 = 111$ series–stratum combinations.

Temporal and seasonal structure. Figure 1 shows the national series. Three regimes are visible: a stable 2014–2018 baseline; the COVID-19 disruption of 2020–2021, when both reporting capacity and clinical demand shifted abruptly; and a post-pandemic regime in 2022–2025 that exceeds pre-pandemic levels. The series exhibits clear intra-annual seasonality, with recurring annual peaks between epidemiological weeks 8 and 20 (February–May).

Spatial, regional, and demographic heterogeneity. State-level case loads vary by more than two orders of magnitude: the highest-volume entities (Ciudad de México, Jalisco, Estado de México, Chihuahua, and Veracruz) together account for about 40% of the cumulative caseload, while sparsely populated southern states

[3] The SINAVE calendar is similar but not identical to the ISO 8601 and CDC Morbidity and Mortality Weekly Report (MMWR) week conventions. We use the calendar as published in the bulletin and do not re-index against either external standard.

are lower but more volatile, which penalizes single-series baselines. Women account for 74.0% of reported cases against 26.0% in men over 2014–2025, exceeding the 1.5–2× female prevalence of major depressive disorder [1]; because the data are *reported incidence*, this likely reflects both clinical patterns and male underreporting. We forecast the three strata separately and revisit the gap in Sect. 6. Based on the indicators available for measurement—a median annual coefficient of variation below 0.25, negligible missing weeks, and a well-contained COVID-19 window—this panel is unusually clean and therefore amenable to data-driven sequence models.

3.2 Proposed Framework

Rationale. The heterogeneity documented above motivates a design that does not commit to one inductive bias. For each series, we train four forecasters whose assumptions are deliberately complementary, and a deterministic rule selects among them. The methodological core is not any individual forecaster—all four are established models—but the principled, per-series choice among them: each series is matched to the inductive bias its own history supports through a deterministic and fully auditable rule, rather than through a global or hand-tuned commitment. A structural additive model captures explicit trend and seasonality and performs well on smooth aggregates; a probabilistic recurrent network with a flexible autoregressive structure performs well on many noisy state series; and two tree-based hybrids add nonlinear interactions among calendar, lag, and demographic features. The four forecasters are exposed through a single, common interface, so that the selection rule operates uniformly across all of them (Fig. 2).

- **Probabilistic recurrent network (DeepAR)** [24,2]: an autoregressive recurrent network that emits a Student-t distribution whose mean is the point forecast. Each federal-entity and regional series is fit with its own DeepAR model; the national aggregate uses a single DeepAR trained jointly over the 32 state series, with the per-state forecasts summed to give the national total. A controlled comparison (Sect. 3.3) shows that joint training adds only a small margin over per-series fitting, so the per-series variant is used at the federal-entity level.
- **Structural additive model (Prophet)** [26]: piecewise-linear trend with automatic change points plus Fourier annual seasonality, fit to the $\log(1+y)$ rate per 100,000 inhabitants to stabilize variance across population sizes.
- **Parallel hybrid (Ensemble)**: an XGBoost regressor [7] using calendar features as well as lagged and rolling-window statistics (lags $\{1, 2, 4, 8, 13, 26, 52\}$ weeks, spanning short-term momentum, quarterly structure, and the annual cycle), blended with Prophet in count space as $\hat{y}_t = \alpha\hat{y}_t^P + (1-\alpha)\hat{y}_t^{XGB}$ with α fit on out-of-fold predictions, so that Prophet's smooth trend and seasonality complement XGBoost's nonlinear lag interactions.
- **Stacked hybrid (Stacking)**: Prophet, ETS and LightGBM [18] experts—two structural/statistical views of trend and seasonality plus one nonlinear

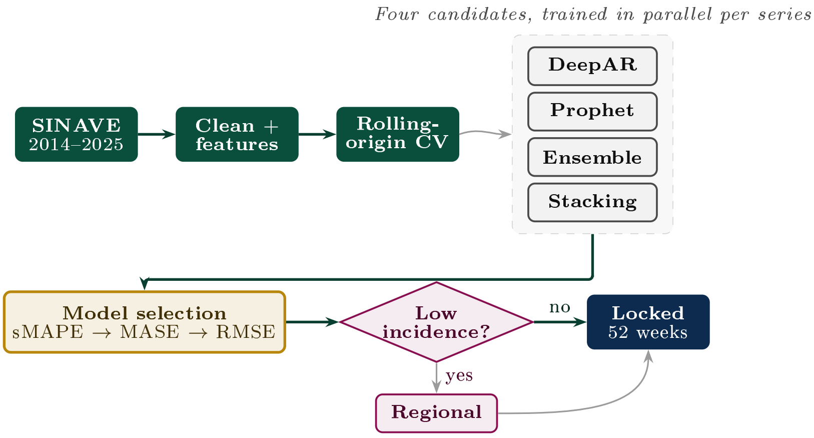

Fig. 2. Proposed framework. Four candidate models with complementary inductive assumptions are trained in parallel per series; a deterministic rule selects one on cross-validated sMAPE (with MASE and RMSE tie-breakers), after which structurally low-incidence series are routed to their regional model before the 52-week forecast is locked.

learner—combined by a nonnegative ElasticNet meta-learner over out-of-fold predictions; the nonnegativity constraint keeps the learned weights interpretable as positive contributions and stable on short series.

Low-incidence regional fallback. Series with structurally low historical incidence—cumulative count below 5 cases in the 52 weeks immediately preceding the forecast origin—are reassigned to the model selected for their socio-urban region (Fig. 2). For such near-zero series, percentage metrics are dominated by noise, so a regional model estimated from a denser signal is a safer default. The rule depends only on past observations and is therefore free of look-ahead bias. In the depression panel studied here, no series falls below the threshold, so the fallback is not activated; it is retained as a safeguard for sparser conditions in the SINAVE catalog.

3.3 Evaluation Protocol

Cross-validation. Every series–stratum combination is evaluated by rolling-origin cross-validation with five folds, a 52-week horizon, and a 13-week (one-quarter) stride; the training window expands at each fold. Metrics are accumulated across folds and weeks, and we report the median across folds per {series, stratum, model} triple.

Metrics and their rationale. Let y_t be the observed value and $\hat{y}_t$ the point forecast at step t over a horizon H. We use four complementary metrics—root mean squared error (RMSE), mean absolute error (MAE), symmetric mean absolute percentage error (sMAPE), and mean absolute scaled error (MASE)—defined in Eqs. (1)–(2), so that no single artifact dominates the selection.

$$RMSE = \sqrt{\tfrac{1}{H}\textstyle\sum_{t=1}^{H}(y_t - \hat{y}_t)^2}, \qquad MAE = \tfrac{1}{H}\textstyle\sum_{t=1}^{H}|y_t - \hat{y}_t|, \tag{1}$$

$$sMAPE = \frac{100}{H}\sum_{t=1}^{H}\frac{|y_t - \hat{y}_t|}{(|y_t| + |\hat{y}_t|)/2}, \qquad MASE = \frac{\frac{1}{H}\sum_{t=1}^{H}|y_t - \hat{y}_t|}{\frac{1}{n-52}\sum_{t=53}^{n}|y_t - y_{t-52}|}. \tag{2}$$

sMAPE is scale-normalized and symmetric, making it comparable across entities with very different volumes, and it is our primary criterion; MASE [16] normalizes by a 52-week seasonal-naive benchmark, so a value below 1 beats seasonal persistence regardless of scale; RMSE penalizes large errors, and MAE is an interpretable case-count error. Excluding the W53 weeks from the two 53-week years shifts every median MASE by less than 0.005 and leaves the ranking unchanged.

Leakage prevention. Temporal leakage is controlled throughout: the rolling-origin splits are strictly expanding; lag and rolling features never reference an observation later than the forecast origin (multi-step rollout fills $h>1$ lags *recursively* with predictions, never future values); demographic covariates are pre-origin annual estimates; and the low-incidence fallback depends only on past counts. No 2026 observation entered the training, tuning or selection of the *frozen* assignment: hyperparameters, architectures and the rule were fixed on data through 2025-W53. The one place where 2026 data drive a choice is the revision experiment of Sect. 5.1, separated in time: reselection sees only W03–W11 and is scored on W12–W18, which it never observes. As a leakage check, an otherwise identical global-versus-local (single-series) DeepAR pair on eight states improved the median sMAPE by only $\sim$5% (24.9 vs. 26.3): a modest, legitimate cross-learning gain, not an information leak. That controlled pair runs a lightweight configuration to isolate the global-versus-local effect, so its absolute errors exceed those of Table 1.

Baselines. Beyond the four candidate models of Sect. 3.2, the 52-week seasonal-naive benchmark is the denominator of MASE and is reported explicitly (Table 1), together with two baselines reimplemented under the identical CV: a tuned Auto-ARIMA with Fourier annual terms and a PatchTST transformer.

Model selection strategy. For a series–stratum combination, let $\mathcal{E}$ be the candidate set and, for each model $e \in \mathcal{E}$, let s_e, a_e, and r_e denote its median cross-validated sMAPE, MASE, and RMSE. Write $s_{(1)} \le s_{(2)}$ for the two smallest sMAPE values and $\mathcal{B} = \{e \in \mathcal{E} : s_e \le 1.05\, s_{(1)}\}$ for the models within 5% of the best sMAPE. The selected model $e^\star$ is determined deterministically:

1. **Primary (sMAPE).** If $s_{(2)} > 1.05\, s_{(1)}$ (the best model beats the second by more than a 5% relative margin), then $e^\star = \arg\min_{e\in\mathcal{E}} s_e$.

2. **Tiebreaker (MASE).** Otherwise $e^\star = \arg\min_{e \in \mathcal{B}} a_e$.
3. **Tiebreaker (RMSE).** If the MASE values in $\mathcal{B}$ are tied, $e^\star = \arg\min_{e \in \mathcal{B}} r_e$.

The low-incidence regional fallback of Sect. 3.2 is then applied. The resulting choice and its metrics are persisted alongside each forecast, so every locked prediction is justified by an auditable record rather than an architectural prior.

Search budget. The per-model search effort is calibrated to its dominant source of sensitivity. Prophet is evaluated using a 12-point grid over the changepoint and seasonality priors, the parameters most sensitive to scale across the 111 panels; DeepAR is fixed at the configuration selected on the validation portion of the first cross-validation fold, with context length and dropout the only series-aware hyperparameters; the gradient-boosting models use the library defaults with the blend weight α fit on out-of-fold residuals. We do not equalize the number of configurations across models, because the marginal value of additional DeepAR tuning at the panel scale is empirically small relative to its computational cost under the production cross-validation protocol.

Code availability. The model-selection, evaluation, and analysis code supporting this study is publicly available at https://github.com/IntegradorIMSS2026Team01/per-series-model-selection2026.

4 Results

4.1 Aggregate Performance

Table 1 reports median per-model accuracy and two coverage indicators across the 111 combinations: the share for which a model beats the seasonal-naive baseline (MASE < 1) and the share for which the selection strategy locks that model as the forecast. These cross-validation figures drive *selection*; their out-of-sample counterpart, examined below and in Sect. 5, is substantially weaker, so they should be read as a selection signal and not as expected live accuracy.

DeepAR achieves the lowest median value for all four metrics by a wide margin and beats the seasonal-naive baseline on 99.1% of combinations. Under the published rule it is selected for 107 of 111 combinations (96.4%), including all $32 \times 3 = 96$ federal-entity combinations; the four exceptions are aggregate series: the three national strata, discussed next, and one sparse southern region, where the two leading candidates fall inside the 5% band: the published rule breaks the tie on MASE and selects Stacking, whereas the frozen baseline assignment of Sect. 5.1 broke it on RMSE and carries DeepAR, hence 108 there against 107 here. DeepAR's federal-entity margin far exceeds the 5% band, so the assignment is insensitive to the exact threshold.

The two discarded candidates of Sect. 2.2—a tuned Auto-ARIMA with Fourier seasonality and a PatchTST transformer—land close to the statistical models and well above DeepAR, so the gap is not an artifact of a weak classical baseline. A grid search over the transformer's patch length and context window does not narrow this gap: across five configurations its general-stratum cross-validated sMAPE stays within 22–23% (best 22.3%), several times higher than the recurrent model's median.

Table 1. Aggregate *cross-validation* performance over the 111 series–stratum combinations (in-sample model-selection metrics, not out-of-sample accuracy; see the generalization note in the text). The first four numeric columns are the *median* of each accuracy metric across the 111 combinations (lower is better). *Beats naive* is the share with MASE < 1; *Selected* is the share locked by the strategy of Sect. 3.3. The two rates do not sum to 100% by row because every model is evaluated on every combination but only one is selected per combination.

	Median accuracy (lower is better)				**Coverage (% of 111)**	
Model	**sMAPE (%)**	**MASE**	**RMSE**	**MAE**	**Beats naive**	**Selected**
Seasonal naive (52-wk lag)	29.28	1.000	22.57	17.87	—	—
DeepAR	**5.99**	**0.171**	**4.79**	**2.19**	**99.1**	**96.4**
Prophet	29.71	0.885	15.95	12.63	68.5	0.9
Ensemble	25.07	0.925	16.37	12.97	62.2	1.8
Stacking	27.56	0.940	17.02	13.75	55.9	0.9
Additional baselines, reimplemented under the identical CV[‡]						
Auto-ARIMA + Fourier	29.07	1.438	—	—	27.0	—
PatchTST (transformer)	27.18	1.409	—	—	27.0	—

[‡] The Auto-ARIMA baseline models $\log(1+y)$ with an ARIMA order selected by AIC over a compact grid and Fourier annual terms (K=4); PatchTST [22] is run with its reference configuration (patch length 16) on the same 104-week context window as DeepAR. Both are deliberately out-of-the-box implementations of widely cited baselines, not exhaustively tuned competitors. We report their scale-free sMAPE and MASE, which cross-check against the production pipeline at the national stratum (sMAPE 8.9% vs. 8.8%; MASE 0.60 vs. 0.59); RMSE/MAE are omitted as their absolute scale is not comparable across the two implementations.

Why the recurrent model dominates in cross-validation, and where it does not. At the federal-entity level, where each series is fit with its own model, two factors favor DeepAR: the flexible autoregressive recurrent structure, combined with a long context window and a Student-t likelihood, captures each state's seasonality and post-pandemic level shift and converges to a strong in-sample fit on this dense, clean panel; and the probabilistic likelihood regularizes training on asymmetric counts. We caution that this is an in-sample cross-validation result that does not persist out of sample (Sect. 5). At the national-aggregate level the advantage is neutralized: the DeepAR forecast is the joint model summed over states, and aggregation smooths the idiosyncratic state variation, so simpler models fit the smooth trend at least as well. The rule therefore locks Ensemble for the national general and women strata and Prophet for men, with cross-validated sMAPE of 8.75%, 7.11%, and 12.64% (MASE 0.59, 0.46, 0.76). That it lands on different models across the three national strata is the per-series mechanism working as intended: each stratum follows its own evidence rather than inheriting the federal-entity winner, so these choices reflect a different data regime, not a contradiction.

Statistical significance. A one-sided Wilcoxon signed-rank test on the 111 paired per-series sMAPE values confirms that DeepAR has lower errors than each of the other three candidates in cross-validation ($p < 0.001$ against Prophet, Ensemble, and Stacking; n=111), so the cross-validation ranking is not an artifact of the median statistic.

Generalization to live 2026 data. Table 1 reports in-sample cross-validation metrics, to be read as the basis for *selection*, not as out-of-sample accuracy. Rescored against the 2026 bulletins the per-series median sMAPE rises for *every* model and the cross-validation gap closes, reflecting both genuine one-year-ahead extrapolation difficulty and the instability of sMAPE on low-count series; Sect. 5.1 shows that the ordering does not survive either. The framework is designed for this: per-series assignments are revisable. Where counts are large and sMAPE is stable (the national strata) the locked forecast generalizes well, as the prospective evaluation quantifies (7.40% over W02–W18); we treat that, not the cross-validation medians, as the primary out-of-sample evidence of this study.

4.2 State-Level Behavior

Across the 32 federal entities, the selected model (DeepAR) ranges from a best case near 1% sMAPE (Michoacán 1.03, Morelos 1.40, Guanajuato 1.94, Baja California 2.09, Coahuila 2.15) to a worst case in the low double digits (Nuevo León 7.91, Querétaro 9.75, Tlaxcala 11.47, Durango 12.91, Puebla 16.28), with MASE below 0.5 throughout, indicating a uniform improvement over seasonal persistence. In the worst-sMAPE case (Puebla, where MASE remains 0.42), the abrupt 2023 level shift inflates the residual variance, but the seasonal phase is still preserved. The per-state out-of-sample errors for every entity, on which the production rule reselects, are released with the source code as supplementary material.

4.3 Demographic Forecasts

Forecasting the three strata separately reveals sex-specific high-demand windows. The selected national female model (Ensemble, sMAPE = 7.11%) projects 116,047 cases for 2026, against 43,500 male cases and 155,966 in the general stratum; the annual peak falls in February–May for both sexes. Because the strata are forecast independently rather than reconciled, the sum of the women and men forecasts exceeds the general forecast by about 2.3%; we treat this small incoherence explicitly and return to reconciliation in Sect. 6.

5 Out-of-Sample Evaluation (2026)

We assess the locked national-general forecast (Ensemble) against every 2026 SINAVE bulletin published since the training cutoff, covering epidemiological weeks W02–W18 (Table 2, Fig. 3). The assignment scored here was frozen before any 2026 observation existed (Sect. 3.3).

Table 2. Out-of-sample evaluation: SINAVE-published weekly incidence (national, general stratum) versus the locked Ensemble forecast for 2026 weeks W02–W18. Deviation is $100(\hat{y} - y)/y$. Forecasts are scored in bulletin week numbering; W01 is not evaluable, as the locked horizon provides no forecast for it, and the bulletin's first week is in any case structurally incomplete (holiday-season carry-over from the prior W52). The 'Recon.' column is the minimum-trace reconciled national-general forecast (Sect. 6).

Week	Obs.	Pred.	Recon.	Dev. (%)	Week	Obs.	Pred.	Recon.	Dev. (%)
W02	2,220	2,022	2,078	−8.9	W11	3,052	3,132	3,190	+2.6
W03	2,819	2,511	2,487	−10.9	W12	2,574	3,004	3,060	+16.7
W04	2,820	2,869	2,834	+1.7	W13	3,223	2,891	2,954	−10.3
W05	2,595	3,044	3,030	+17.3	W14	2,341	2,897	2,955	+23.8
W06	2,585	2,973	3,012	+15.0	W15	2,981	2,961	3,030	−0.7
W07	2,975	2,886	2,961	−3.0	W16	3,065	3,093	3,124	+0.9
W08	3,041	2,906	2,997	−4.4	W17	3,075	3,048	3,112	−0.9
W09	3,086	3,046	3,131	−1.3	W18	2,791	3,032	3,080	+8.6
W10	3,057	3,167	3,227	+3.6					

Cumulative W02–W18: observed 48,300, predicted 49,482 (reconciled 50,261), deviation +**2.45**% (reconciled +4.1%). Aggregates are computed on unrounded values, so a column of rounded cells may differ by one case.

Aggregate behavior. Over the 17 complete weeks W02–W18, the locked forecast attains an sMAPE of **7.40%** (mean MAE = 205 cases per week) and a cumulative deviation of +2.45%, within the ±5% planning tolerance. The relevant comparator is the CV-period sMAPE for the same national-general stratum, 8.75%: the out-of-sample error is below the in-sample CV error. The weekly errors have a median absolute percentage error of 4.4%, below the 6.6% median of the per-week CV errors, so the forecast generalizes without visible drift. The ranking is robust to the error metric: under a count-appropriate mean Poisson deviance, the locked Ensemble forecast is again the best of the four models in this window (25.3, against 30.4 for Prophet, 57.9 for DeepAR, and 104.9 for Stacking). Uncertainty, however, does not transport as well as the point forecast. The empirical 80% residual band, calibrated from the 10th and 90th percentiles of post-2021 residuals, covered 10 of the 17 complete prospective weeks (58.8%). This undercoverage indicates that the historical residual distribution did not transport to

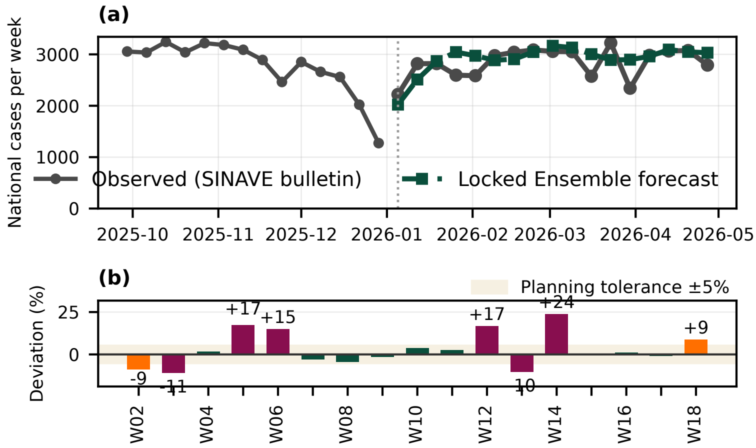

Fig. 3. 2026 out-of-sample evaluation, national general stratum. (a) Late-2025 context and W02–W18 observed (gray) versus the locked Ensemble forecast (green), scored in bulletin week numbering. (b) Per-week deviation against the ±5% planning-tolerance band (shaded), with bars colored by absolute deviation.

W02–W18; the band should therefore be interpreted as a diagnostic uncertainty summary rather than as calibrated prospective coverage.

Pairwise comparison. We compare the locked Ensemble against each rival with a Diebold–Mariano test carrying the Harvey–Leybourne–Newbold small-sample correction, on the W02–W18 squared-error losses. The unadjusted two-sided p-values are 0.0167 against DeepAR, 0.0178 against Stacking, and 0.4328 against Prophet. These are three exploratory comparisons on the same seventeen weeks, so we adjust them: under Holm the first two both become $p = 0.0502$ and neither clears the conventional 5% threshold. The losses also come from a single frozen multi-horizon trajectory, so serial dependence may not be fully absorbed at h=1. We therefore read the result as indicative of lower squared-error loss against DeepAR and Stacking, not as an established difference, and as no evidence at all against Prophet. With seventeen weeks the test has little power, so a non-rejection means insufficient evidence to separate the leading models rather than equivalence.

Largest single-week deviation. W14 has the largest single-week deviation (+23.8%), well beyond the upper quartile of the cross-validation error band. The value remained at 2,341 cases through the manuscript cutoff and is therefore not a reporting-lag artifact; the four subsequent weeks, W15–W18, return to an sMAPE of 2.7%, so the miss does not propagate into the trajectory. We treat W14 as one observation in the tail of the in-sample error distribution rather than as evidence of model drift.

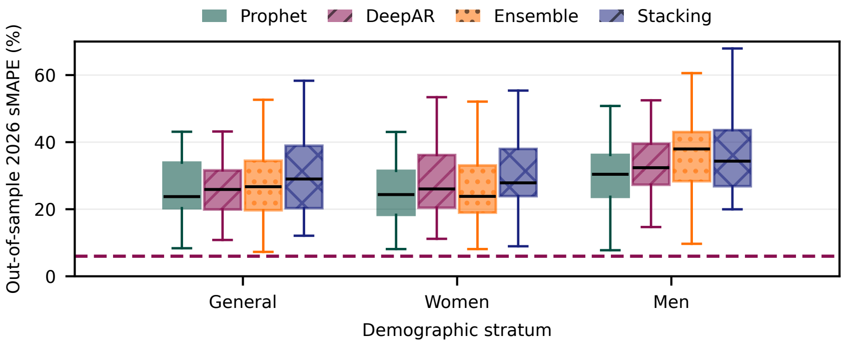

Fig. 4. Out-of-sample 2026 per-series symmetric error by model and demographic stratum over W03–W18, the weeks common to every series; each box pools the 32 federal-entity series with the national series for that stratum (n=99). In cross-validation the recurrent model is selected for most series at a 5.99% median (dashed line); out of sample that advantage disappears and the four models converge, with per-series medians of 26.7% for Prophet, 27.4% for DeepAR, 28.1% for Ensemble, and 29.3% for Stacking, worst on the sparse male stratum. Treating these scores as a revisable signal reassigns 55 of the 99 series (62 of 111 when the derived regional series are included).

Per-series behavior. Cross-validation accuracy does not persist per series out of sample (Sect. 4); Fig. 4 resolves this by model and demographic stratum. The cross-validated dominance of the recurrent model (dashed line) does not survive: out of sample no model is uniformly best, and the error grows toward the sparse male stratum, where low weekly counts also inflate sMAPE because of small denominators—on the scale-free MASE, scored against 52-week seasonal persistence, the same male series sit between 0.74 and 0.90 across the four models: better than persistence, but by a far narrower margin than the cross-validated figures suggest. This is the regime in which an *auditable, revisable* rule earns its keep, and Sect. 5.1 tests that claim directly against the frozen baseline assignment.

5.1 Ablation of the Candidate Pool

A four-model pool invites a direct question: how much of the result depends on having four? We re-run the selection rule restricted to each subset of the pool and rescore the resulting assignments without refitting—the rule adjudicates among forecasts that were already locked. Two regimes are separated: *static*, selecting once from cross-validated evidence as deployed, and *dynamic*, reselecting on W03–W11 and scoring on the later, unused weeks W12–W18. Selection runs over all 111 series; scoring runs over the 99 with a direct bulletin observation, the 12 derived regional series being a sensitivity. Policies are compared pairwise over the same series (Wilcoxon signed-rank), since comparing each policy's own reassigned subset would compare different sets.

Table 3. Ablation of the candidate pool: median out-of-sample sMAPE (%) over the 99 directly observed series, W03–W18. *Static* selects once from cross-validated evidence; *dynamic* reselects on W03–W11 and is scored on the unused weeks. p: two-sided Wilcoxon signed-rank against the full pool over the same series. The frozen baseline is the assignment locked before 2026, and what the dynamic regime must beat.

	Static		**Dynamic**	
Candidate pool	sMAPE	p	sMAPE	p
Full pool (four models)	27.4	—	28.5	—
without Prophet	27.4	0.84	29.4	0.07
without DeepAR	28.0	0.49	26.8	0.03
without Ensemble	27.4	0.69	28.8	0.17
without Stacking	27.4	1.00	28.8	0.72
Prophet alone	26.7	0.09	27.6	0.99
Frozen baseline assignment	—		32.7	—

The pool is not what carries the result. Removing any single candidate leaves the static median essentially unmoved: the four leave-one-out variants fall within 0.6 points of the full pool and none approaches significance (Table 3). This holds even when the removal reorganizes the assignment completely—dropping DeepAR moves 96 of the 99 scored selections onto other models and moves the median by 0.6 points. Against a single-model policy the comparison is no more favorable: Prophet alone attains the best static point estimate, and we find no evidence that the full selection beats it (p=0.09; p=0.07 on the 111-series sensitivity). We therefore do not claim that the candidate pool improves accuracy on this panel.

The cross-validated ranking does not transport. The direction of the discrepancy matters more than its size. Prophet is the *worst*-ranked candidate in cross-validation (Table 1) and the best in the prospective window. The evidence the rule selects on is thus not merely noisy out of sample; on this panel it does not carry the ordering it was selected for. This is the same phenomenon quantified in Sect. 4, isolated to the selection mechanism itself.

What does hold is revision. The dynamic column separates the two claims. Against the frozen baseline—the assignment locked before 2026—reselecting on prior weeks lowers the median from 32.7% to 28.5% on weeks the reselection never saw, and every pool tested improves on that baseline. The gain comes from revising, not from the pool: the full pool does not beat Prophet alone here either (p=0.99). The best leave-one-out variant in this regime, the pool without DeepAR (26.8%, p=0.03), is a comparison selected after seeing the outcome, and we do not propose it as a superior configuration. The framework's value on this panel is therefore not that four models beat one; it is that a mechanism with an auditable record detects when a frozen assignment has stopped transporting and revises it using only observations preceding the evaluated weeks. A negative ablation is what an auditable design is supposed to surface, and it bounds how far a cross-validated selection should be trusted once the signal moves.

6 Discussion

Two findings should be read together. The ablation of Sect. 5.1 does not support the claim that the candidate pool improves accuracy here: no leave-one-out variant separates from the full pool, and a single-model policy matches it. What the same experiment does support is revision. Against the frozen baseline assignment, reselecting on prior weeks lowers the median error on weeks the reselection never saw, and the auditable log is what makes that possible: it records the evidence behind each choice, so a selection that has stopped transporting can be shown to have done so and replaced on stated grounds rather than on an architectural preference.

Public-health relevance. The 74%/26% reported female–male split, broadly echoed by the 2026 forecast (about 73%/27%), has planning implications: a service grid sized to the national mean may underserve women in absolute terms, while male depression remains plausibly under-identified. A sex-disaggregated 52-week forecast could support both proactive capacity allocation for the February–May saturation window and male-targeted help-seeking campaigns in quieter months. We emphasize that the data represent *reported incidence*, not prevalence; the female–male gap is a hypothesis requiring dedicated epidemiological validation before being used as policy input. This paper evaluates the framework; it does not describe a deployed operational system.

Hierarchical reconciliation. The three demographic strata are forecast independently, leaving the $\sim$2.3% general-versus-(women+men) incoherence noted in Sect. 4.3. Applying a minimum-trace reconciliation [27] to the locked national forecasts, weighted by the per-stratum cross-validation error variances, removes the incoherence by construction; the reconciled national-general series (Table 2, 'Recon.' column) shifts the operational forecast by only 1.9% on average and, because it is already well calibrated, slightly raises the cumulative deviation (+4.1% vs. +2.45%). The independence approximation is thus a small source of error, and we report the unreconciled forecasts for transparency.

Limitations. Seven limitations constrain the conclusions. (i) Only one condition (depression) is evaluated; transfer to conditions with different temporal behavior remains to be demonstrated. (ii) The per-series cross-validation metrics of Table 1 are optimistic relative to live performance, and Sect. 5.1 shows the ranking does not transport. The out-of-sample claim therefore rests on the national-aggregate prospective evaluation and on the revision experiment, not on the cross-validation medians. The search budget is also not equalized (Sect. 3.3): Prophet, the candidate given the broadest explicit search, has the lowest out-of-sample point estimate, but the differences are not significant and cannot be attributed causally to tuning effort. (iii) The headline prospective evaluation is at the national-aggregate stratum over an early-2026 window (W02–W18); the per-series analysis runs over W03–W18, the sixteen weeks for which every series has both a bulletin observation and a locked forecast, and the per-state table is released with the source code, but a per-state *full-year* evaluation is still pending. (iv) The most recent bulletins are subject to reporting-lag revision, so the latest one or two weeks are provisional. (v) Revising point-forecast assignments does not by itself calibrate predictive uncertainty: rolling recalibration or direct probabilistic evaluation is required, as the undercoverage of the empirical band in Sect. 5 shows. (vi) The framework has not been externally validated outside SINAVE/Mexico. (vii) SINAVE is a passive, facility-based system, so the panel measures *reported* incidence: it counts episodes that reached care and were coded, not episodes that occurred. Two consequences bear on generalizability. Diagnostic and reporting capacity differs across federal entities, so part of the level difference between states reflects surveillance capacity rather than epidemiology; and care-seeking differs by sex, which is the most likely source of the male shortfall noted in Sect. 3.1. A model fitted to these series therefore learns the reporting process together with the disease process, and the selection rule inherits both. This does not invalidate the forecasts for planning—services are sized against reported demand, which is what the panel measures—but it does mean the framework should not be read as forecasting community-level incidence, and that transfer to a system with different case-ascertainment rules must be demonstrated, not assumed.

Future work. We plan to extend the framework to additional CIE-10 codes in the SINAVE catalog, to report a full-year per-state prospective evaluation, to add tuned SARIMA and N-BEATS/N-HiTS baselines, and to pursue external validation. A further direction is to make the *forecasts* explainable, not only the selection. The rule explains which model was chosen and on what evidence, but not why that model expects a given trajectory; post-hoc attribution over the tree-based hybrids and saliency over the recurrent model's context window would let a planner see which lags and calendar effects drive a specific weekly figure. That is a different guarantee from the one offered here, and we treat it as complementary rather than as a refinement of the rule.

7 Conclusions

We presented an empirical study of multi-model, per-series forecast selection for weekly depression incidence at federal-entity granularity in Mexico, comparing a structural model, a global probabilistic recurrent network, and two tree-based hybrids under a deterministic, auditable, and revisable selection rule. Cross-validation on more than a decade of SINAVE data, across a nationwide panel of series and demographic strata, selects the recurrent model (DeepAR) for the large majority of federal-entity series, while the rule retains simpler models where aggregation favors them. We explicitly treat this ranking as an in-sample selection signal: on live 2026 data the advantage narrows, and an ablation finds no evidence that the pool beats a single-model policy. The contribution is therefore not that several models beat one, but that an auditable mechanism detects when a frozen assignment has stopped transporting and revises it using only prior observations. We are not aware of prior work that applies a multi-model framework with auditable, revisable per-series model selection to psychiatric surveillance at federal-entity granularity in a Latin American country. The framework is a research prototype.

Ethics and Data Availability

The SINAVE bulletin reports aggregate weekly case counts at the federal-entity level, stratified by sex. All analyses were performed on these aggregate cells, and no cell corresponds to an individual patient; therefore, no informed consent or review-board approval was required. The data are published as open data by the Dirección General de Epidemiología, Secretaría de Salud, and were retrieved from the public SINAVE portal on 2026-04-15 [11]; redistribution is governed by the bulletin's open-data terms.

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A Implementation and Reproducibility Details

The four models share a single interface, configured from hierarchical YAML files under OmegaConf. The framework runs on Python 3.12 with GluonTS 0.15 (PyTorch 2.5), Prophet 1.1, XGBoost 2.0, LightGBM 4.3, scikit-learn 1.4 and pandas 2.2; DeepAR trains on one NVIDIA T4 GPU and the rest on CPU. Every seed source is fixed at 42 and cuDNN runs deterministically. Table 4 lists the per-model hyperparameters; the configuration files, the pseudocode and the per-state table are in the repository cited in Sect. 3.3.

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Table 4. Principal hyperparameters, fixed before the 2026 evaluation.

Model	Principal hyperparameters
DeepAR	2 layers $\times$ 80 cells; context 104 wk; horizon 52; dropout 0.15; Student-t output; Adam η=5$\times$10^{-4}, λ_2=10^{-6}; batch 32; $\leq$300 epochs; early stop (patience 15).
Prophet	rate per 100k, log(1+y); changepoint prior $\in \{0.01, 0.03, 0.05\}$; seasonality prior $\in \{0.025, 0.05, 0.1, 0.5\}$ (12-point grid under temporal CV).
Ensemble	XGBoost on calendar/lags $\{1, 2, 4, 8, 13, 26, 52\}$/rolling stats; blend weight learned on out-of-fold predictions.
Stacking	Prophet + ETS + LightGBM experts; nonnegative ElasticNet (α=1.0, `l1_ratio`=0.5).

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